HALO: Human-Aligned Data Collection for WhoLe-body MObile Manipulation

Author One[1], Author Two[1], Author Three[1], Author Four[2], Author Five[1,*], and Author Six[1,*]

[1]Department Name, University Name, City, Country.

[2]Department Name, Another University, City, Country.

[*]Equal Advising.

Emails: {author1, author2, author3, author5, author6}@university.edu, author4@another.edu

Learning whole-body mobile manipulation requires demonstrations that coordinate bimanual actions with locomotion across extended workspaces. However, existing interfaces trade off direct robot control, perception and action alignment, and natural operator mobility, making such demonstrations difficult to collect. We introduce HALO, a human-aligned data collection system for whole-body mobile manipulation. The system aligns the operator's perception and action space with the robot while preserving direct physical feedback and natural locomotion. Operators guide the robot end-effectors through kinesthetic teaching and move a low-resistance platform without the deployed mobile base while performing bimanual manipulation. This interface enables more natural whole-body demonstrations, but the close human–robot coupling introduces discrepancies between the demonstration and deployment settings. We address these discrepancies with deployment-aligned demonstration processing, which calibrates and temporally aligns platform motion with the deployed base response and masks and inpaints demonstrator hands in recorded observations. Compared with Mobile ALOHA and VR teleoperation, HALO enables up to 2.9× faster demonstration collection while maintaining comparable or higher success rates. Policies trained with HALO perform precise manipulation tasks, including zip-tie fastening, and achieve a 73.3% success rate on a six-stage long-horizon mobile manipulation task involving repeated coordinated base and arm motion.

1. Introduction

End-to-end robot policies and imitation learning (Zhao et al., 2023; Chi et al., 2025; Kim et al., 2024; Black et al., 2024; Li et al., 2026; Wu et al., 2026) have achieved strong performance across a wide range of real-world manipulation tasks. However, most prior work focuses on tabletop settings, where robots are confined to a fixed workspace. Real-world applications instead require robots to extend their workspace through mobility while continuing to manipulate objects, motivating coordinated whole-body bimanual manipulation (Fu et al., 2024). Learning such behaviors requires efficient collection of demonstrations that jointly capture locomotion and manipulation.

Existing approaches to whole-body data collection make different trade-offs in preserving natural human coordination. Indirect teleoperation methods (e.g., VR-based interfaces (Zhu et al., 2022; Rakita et al., 2018; Honerkamp et al., 2025)) provide less direct manipulation control and lack force feedback. Leader–follower systems (e.g., Mobile ALOHA (Fu et al., 2024)) provide more direct manipulation through leader arms and kinesthetic teaching, but the spatial separation between the operator's hands, viewpoint, and robot end-effectors introduces a perception gap and constrains natural operator mobility.

Retargeting-based approaches (e.g., UMI (Chi et al., 2024) and HoMMI (Xu et al., 2026)) instead preserve unconstrained human motion by mapping human demonstrations to robot behavior through retargeting or whole-body control. Taken together, existing approaches balance direct robot control, perception–action alignment, and natural operator mobility in different ways, often emphasizing one property over another. These trade-offs across perception, manipulation, and mobility are summarized in Tab. 1.

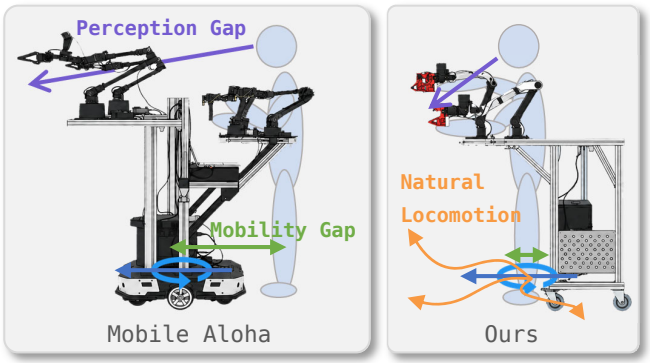


Figure 1: **Interface comparison.** Mobile ALOHA exhibits perception and mobility gaps, while our interface aligns human–robot perception and preserves natural operator mobility.

Effective whole-body demonstration requires preserving the robot's native action constraints and direct physical feedback while **aligning the operator's perception–action space** with the robot and **preserving the operator's natural mobility**, as illustrated in Fig. 1. We adopt two design principles: First, the manipulation interface should preserve direct robot control and feedback. We therefore retain the robot arms and use kinesthetic teaching (Akgun et al., 2012), keeping the operator close to the end-effectors to reduce the **perception gap** while preserving direct visual and force feedback. Second, the mobility interface should preserve the operator's natural locomotion without requiring control of the mobile base. We therefore remove the mobile base during demonstration, reducing the **mobility gap** and allowing natural body motion for repositioning.

While these interfaces enable more natural whole-body demonstrations, the resulting human–robot coupling introduces visual and control distribution shifts between demonstration data and autonomous deployment (Xu et al., 2026). We identify these discrepancies and introduce a deployment aligned post-processing pipeline that removes demonstrator-induced artifacts from both visual observations and control signals, aligning the collected demonstrations with policy execution.

Together, these design choices improve demonstration efficiency and whole-body coordination. Compared with Mobile ALOHA and VR teleoperation, HALO enables up to 2.9× faster demonstration collection while maintaining higher success rates. The resulting policies achieve sub-millimeter manipulation precision and a 73.3% success rate on a six-stage long-horizon mobile manipulation task involving repeated coordinated base and arm motion. These results demonstrate that human-aligned data collection supports efficient, dexterous, and coordinated whole-body policy learning. Our contributions are threefold:

- **Human-aligned data collection system.** We introduce a whole-body bimanual teaching system that aligns the operator's perception–action space with the robot while preserving natural mobility, enabling efficient collection of coordinated and dexterous mobile manipulation.
- **Deployment-aligned data post-process.** We identify and mitigate demonstrator-induced artifacts in visual observations and control, enabling demonstrations collected under tight human–robot coupling to be used for autonomous policy learning.
- **Systematic real-world evaluation.** We evaluate the proposed system in terms of data-collection efficiency, demonstration success, and further study the effects of different inference strategies as well as the achievable precision and efficiency of the learned policies.

2. Related Works

2.1. Robot Demonstration Collection

Traditional teleoperation minimizes the embodiment gap by directly controlling the target robot, but interfaces such as SpaceMouse devices, joysticks, and VR controllers remain indirect (Zhu et al., 2022;

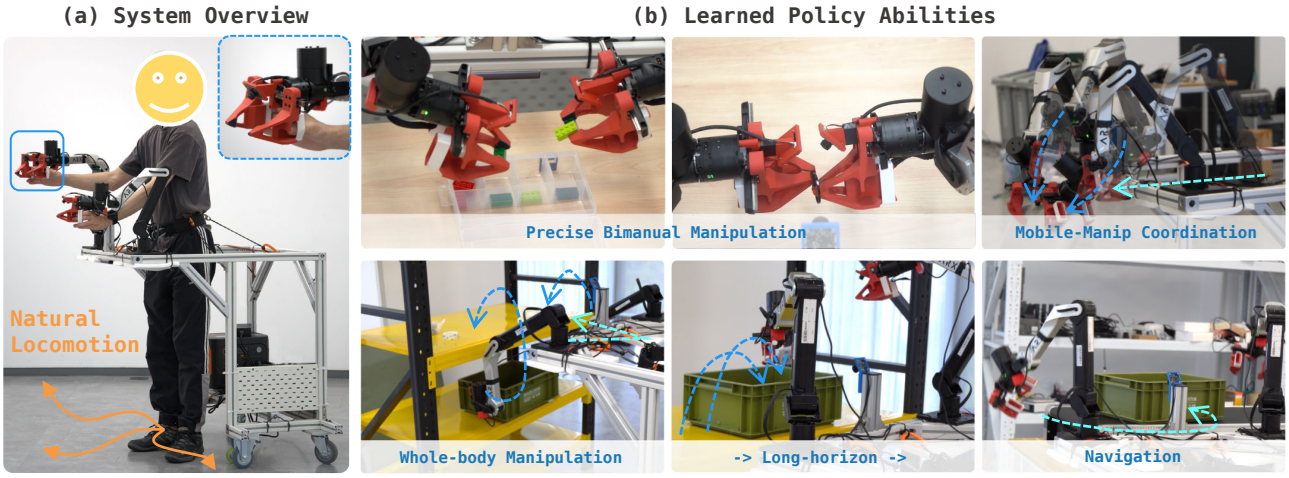


Figure 2: **HALO overview and learned capabilities.** (a) Human-aligned data collection with natural locomotion and direct bimanual kinesthetic teaching. (b) Learned policies exhibit precise manipulation, coordinated whole-body control, long-horizon mobile manipulation, and navigation.

Table 1: **Qualitative comparison of data collection systems.** We compare representative interfaces in terms of the desired properties for whole-body mobile manipulation data collection.

Method	Mobile Support	Egocentric Perception	Force Feedback	Natural Locomotion	Retargeting-Free Actions	Precise Base Control
GELLO (Wu et al., 2024b)	×	×	×	×	✓	×
Mobile ALOHA (Fu et al., 2024)	✓	×	×	×	✓	✓
VR (Zhu et al., 2022; Rakita et al., 2018; Honerkamp et al., 2025)	✓	×	×	×	✓	✓
Exoskeleton (Citron et al., 2026; Liang et al., 2026)	✓	✓	✓	✓	×	×
UMI (Chi et al., 2024) / HoMMI (Xu et al., 2026)	✓	✓	✓	✓	×	×
HALO	✓	✓	✓	✓	✓	✓

Rakita et al., 2018; Honerkamp et al., 2025; Wong et al., 2022). Kinematic replicas, such as ALOHA (Zhao et al., 2023; Fu et al., 2024), GELLO (Wu et al., 2024b; Jiang et al., 2025), and exoskeleton-based approaches (Citron et al., 2026; Chen et al., 2026), provide more intuitive control but typically lack force feedback, although some recent systems incorporate it (Liang et al., 2026); they also become cumbersome in mobile settings. UMI (Chi et al., 2024) enables scalable in-the-wild data collection, while recent extensions (Xu et al., 2026; Yu et al., 2025) incorporate egocentric sensing and whole-body controllers for mobile manipulation; however, base motion remains difficult to control precisely, and visual and control gaps persist. Our system instead aligns the operator with the robot, providing direct visual and force feedback.

2.2. Human-to-Robot Demonstration Transfer

Transferring human demonstrations to robot policy learning requires bridging the visual and action gaps between human embodiment and robot execution. DexUMI (Xu et al., 2025) removes the human hand through inpainting and uses action replay to obtain the corresponding foreground dexterous robot hand.

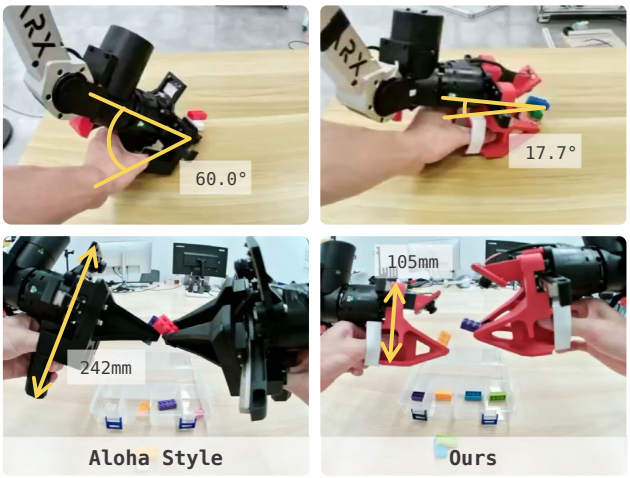


Figure 3: **Kinesthetic end-effector redesign.** Compared with the ALOHA end-effector, we remove the handle-and-scissor design for kinesthetic teaching. The handle limits access near the table surface, while the scissor mechanism is less convenient for forceful manipulation. Our design increases hand contact area, enabling compact and stable physical control.

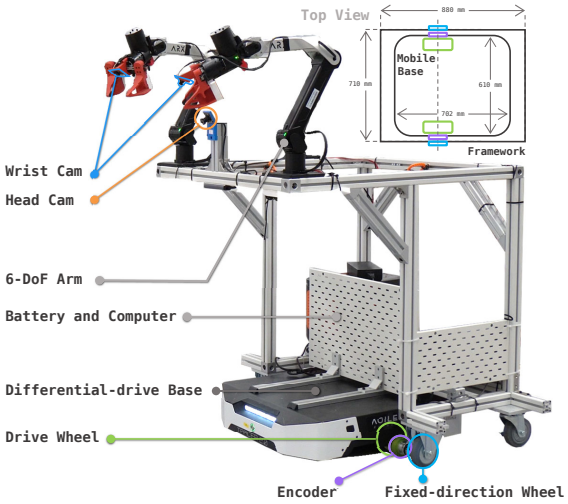


Figure 4: **HALO deployment system overview.** Hardware configuration includes the dual-arm manipulation interface, onboard cameras, differential-drive base, encoders, and computing system. The base is removed during data collection.

EgoBridge (Punamiya et al., 2025) aligns human and robot demonstrations using dynamic time warping. Ego2Robot (Wang et al., 2026) first estimates the human hand pose and then replaces the hand with the robot arm through inpainting. Human2Sim2Robot (Lum et al., 2025) uses a single human demonstration together with reinforcement learning to derive robot-executable actions. In contrast, since our kinesthetic teaching interface directly provides ground-truth robot actions, we only need to mitigate demonstrator-induced artifacts in visual observations and control, without additional action annotation or replay.

2.3. Learning-Based Mobile Manipulation

Hierarchical models separate task-level manipulation prediction from whole-body execution, using planners or whole-body controllers to coordinate the base and arm (Zhu et al., 2026; Ha et al., 2024; Yuan et al., 2026; Xu et al., 2026; Zhao et al., 2025). However, many real-world tasks require tight and continuous coordination between mobility and manipulation, making such modular designs less suitable for unified whole-body control and introducing additional system complexity. Unified policies (Wu et al., 2024a; Intelligence et al., 2025; Fu et al., 2024), in contrast, directly model coupled base and manipulation actions, which is particularly effective when mobility and manipulation must evolve simultaneously. We find that simply concatenating base actions with arm actions already yields strong whole-body control without additional architectural design.

3. Method

HALO combines kinesthetic teaching with a low-resistance mobility interface to collect coordinated whole-body demonstrations. We first describe the manipulation interface in Section 3.1 and the mobility interface in Section 3.2. We then introduce deployment-aligned demonstration processing in Section 3.3, including visual processing and base-motion alignment, to address the discrepancies caused by close human–robot coupling and base-free data collection. Finally, we describe policy learning and asynchronous execution in Section 3.4.

3.1. Manipulation Interface

A data collection interface should adapt to the operator rather than require the operator to adapt to the robot. ALOHA's leader–follower design enables intuitive bimanual control, but the operator's hands remain spatially separated from the follower end-effectors, and the operator's egocentric view differs from the robot viewpoint. These gaps weaken perception–action alignment and direct robot control. We therefore use kinesthetic teaching to more closely align the operator's perception and action space with the robot.

As shown in Fig. 3, the original ALOHA end-effector is not designed for this type of physical interaction. Its handle limits the working space, while the scissor mechanism is less suitable for forceful manipulation. These features make stable grasping and force application less convenient in compact or contact-rich tasks.

So we redesign the end-effector for kinesthetic teaching by removing the handle-and-scissor mechanism and adopting a compact, high-contact design. As shown in Fig. 3, the redesign reduces the end-effector size by more than half and increases the available pitch angle by about 40°, enabling closer and more flexible manipulation while preserving direct visual and force feedback.

This close human–robot coupling makes bimanual demonstrations more natural, but it also causes the operator's hands to appear in the robot observations. We address this visual discrepancy in the deployment-aligned demonstration processing stage described in Section 3.3.

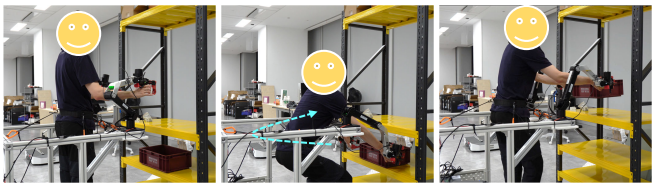


Figure 5: **Implicit base control.** The low-resistance collection platform allows base motion to arise naturally from the operator's movement, without explicit base control that could disrupt hand manipulation, thereby capturing coordinated mobile manipulation behaviors.

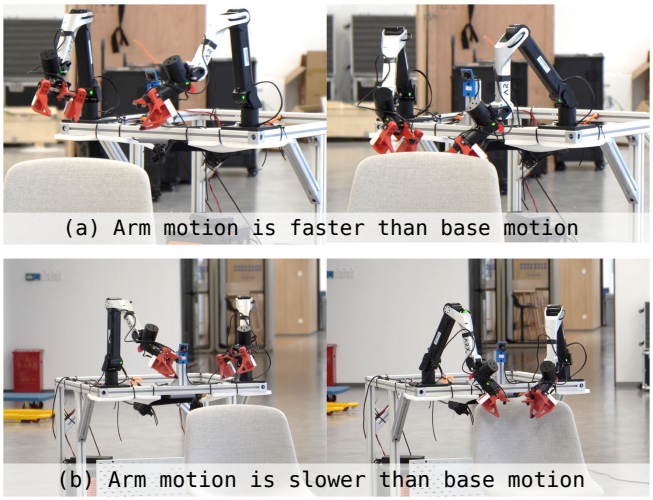


Figure 6: **Failure cases in tasks requiring high mobility.** (a) Without compensating for mobile-base latency, the arm motion advances faster than the base motion. (b) Synchronous inference introduces additional control latency, causing the base to travel farther than intended.

3.2. Mobility Interface

The mobility interface preserves natural operator locomotion while maintaining the motion constraints of the deployed base. Unlike Mobile ALOHA (Fu et al., 2024), whose powered base occupies the operator's movement space, we remove the base during data collection and let natural body motion directly drive a low-resistance demonstration platform, as shown on the left of Fig. 2 and Fig. 5. This allows the operator to control mobility Implicitly while focusing on bimanual manipulation.

The deployment system uses an AgileX Tracer AGV 2 differential-drive base, whereas the collection platform uses passive wheels and two fixed-direction wheels placed along the drive-wheel axis. This configuration mechanically constrains the collection platform to the same nonholonomic motion space as the deployed base, preserving kinematic compatibility without post-hoc retargeting.

Removing the powered base also substantially reduces the force required for repositioning: moving the deployment system requires 47.3 N of horizontal force and Mobile ALOHA requires 13.0 N (Fu et al., 2024), whereas our collection platform requires only 2.3 N. This reduction allows the operator to walk and reposition naturally, as shown in Fig. 5.

Two encoders mounted near the drive-wheel axis measure the motion of the collection platform, as shown in the upper right of Fig. 4. Since the collection platform is passive and has different dynamics from the deployed base, the measured motion cannot be used directly as the deployment action. We therefore calibrate the mapping from platform motion to base velocity and estimate the response delay of the deployed base in Section 3.3.2.

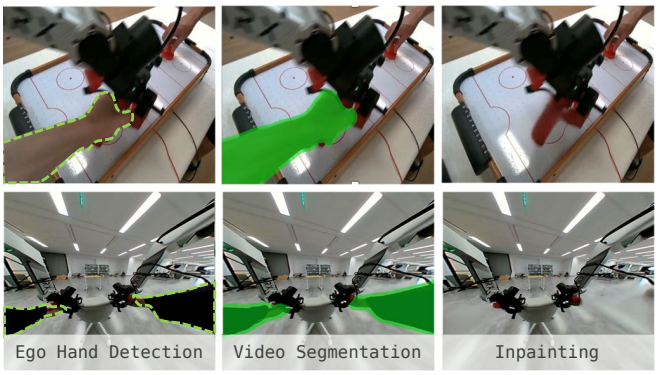


Figure 7: **Visual processing.** Our pipeline removes human hands during rapid motion, bimanual manipulation, and egocentric interaction.

3.3. Deployment-Aligned Demonstration Processing

The close human–robot interaction and base-free collection platform make demonstrations more natural, but they also introduce discrepancies between the demonstration and deployment settings. We address these discrepancies through deployment-aligned demonstration processing, covering both visual observations and base actions.

3.3.1. Visual Processing

The operator's hands appear in the robot observations during kinesthetic teaching. To remove these demonstrator-specific cues while preserving task-relevant scene information, CaRE-Ego (Su et al., 2025) detects the ego operator's hands, SAM2 (Ravi et al., 2025) propagates the masks across frames for temporal consistency, and DiffuEraser (Li et al., 2025) reconstructs the masked regions using an optical-flow prior. The resulting observations better match the visual distribution encountered during autonomous execution, as shown in Fig. 7.

3.3.2. Base Motion Alignment

With the kinematic constraints already matched by the collection interface, the remaining alignment accounts for differences in dynamics and control response between the passive platform and the powered base. Encoder readings are converted into left and right platform wheel velocities, u_L and u_R. We characterize the response of the deployed base from commanded velocities to encoder-observed motion and estimate the resulting action–state delay for temporal alignment, following (Fu et al., 2024). Ignoring this delay introduces a systematic mismatch between commanded actions and observed base motion, as illustrated in Fig. 6. We further fit an affine mapping from the measured platform motion to deployment-compatible base velocities:

$$\begin{bmatrix} v \\ \omega \end{bmatrix} = \begin{bmatrix} k_{vL} & k_{vR} \\ k_{\omega L} & k_{\omega R} \end{bmatrix} \begin{bmatrix} u_L \\ u_R \end{bmatrix} + \begin{bmatrix} d_v \\ d_\omega \end{bmatrix}, \tag{1}$$

where v and ω are the deployment-compatible linear and angular velocities. The coefficients and offsets are estimated from paired platform measurements and commanded base velocities collected on the deployment system. To avoid compounding errors from integrating estimated motion, we use direct base velocity rather than pose as the action representation.

3.4. Policy Learning

To evaluate the effectiveness of the demonstrations collected with HALO, we use the pretrained $\pi_{0.5}$ vision-language-action model (Intelligence et al., 2025) as the base policy and fine-tune it on our demonstrations. The policy receives three images, including one head-camera image and two wrist-camera images, together with the robot joint positions and base linear and angular velocities. The action vector concatenates the arm actions with the base linear and angular velocities. For each task, we fine-tune the pretrained policy with AdamW using a constant learning rate of 5×10^{-5}, a batch size of 128, and 5k training steps. Models trained without pretrained initialization use the same training setup for 10k steps.

Tight whole-body coordination requires continuous policy execution with frequent replanning. We therefore use asynchronous real-time action chunking (RTC) (Black et al., 2025), where the policy predicts the next action chunk while the current chunk is being executed. Given the latest observation o_t and the committed action prefix a_{t+d-1}, the policy predicts the next action chunk:

$$a_{t+d:t+d+H-1} \sim \pi(\cdot \mid o_t, a_{t+d-1}), \tag{2}$$

where d is the inference delay in control steps and H is the prediction horizon. Conditioning on the committed action prefix maintains continuity between consecutive action chunks. We further ablate the effects of inference settings, pretraining, and action representation on policy performance.

4. Evaluations

We evaluate our framework along three key dimensions that reflect its ability to preserve human manipulation skills while enabling efficient and coordinated whole-body policy learning:

- **Data collection efficiency.** How efficiently can users collect successful whole-body demonstrations with our system compared with existing teleoperation interfaces?
- **Dexterity.** How effectively can demonstrations collected with our system transfer human manipulation dexterity to learned robot policies, particularly for precise and bimanual manipulation?
- **Coordinated whole-body control.** How well can the learned policy coordinate base motion and manipulation, especially for tasks requiring precise repositioning and long-horizon execution?

For all experimental tasks, we limit the complete data collection process to one hour, including scene reset and object preparation. We impose no constraints on demonstration variability, such as fixed initial base poses or object placements. Operators are instructed to collect demonstrations at a natural pace and to walk naturally during mobile manipulation, allowing us to evaluate whether HALO can capture natural human whole-body motion.

4.1. Data Collection Efficiency

We compare HALO with VR-based teleoperation and Mobile ALOHA in terms of demonstration success and data collection efficiency, focusing on tasks that require tight coordination between mobility and bimanual manipulation. For a fair comparison, all methods are evaluated by the same operator under the same task settings, collection time budget, and reset protocol. The operator had no prior experience with any of the evaluated interfaces before the study.

Task description. We evaluate two tasks, *Push Chair* and *Disassemble Lego*. In *Push Chair* (Fig. 9), the operator grasps the chair back with both grippers while moving at approximately normal walking speed ($\sim 0.7\,\mathrm{m/s}$), then navigates to push the chair under a table. This task requires stable bimanual grasping during rapid base motion. In *Disassemble Lego* (Fig. 9), the operator disassembles a four-piece Lego structure and places each piece into the bin of the corresponding color. The Lego configuration and object arrangement are randomized, requiring precise and adaptive bimanual manipulation.

Results. HALO matches or exceeds the baselines in demonstration success rate while reducing the average duration of successful demonstrations by up to 66%. Compared with Mobile ALOHA, HALO achieves speedups of 2.9× on *Push Chair* and 2.8× on *Disassemble Lego*, while improving the latter's demonstration success rate by 20 percentage points. As illustrated in Fig. 8, the two baselines present distinct interaction difficulties: VR's perception gap hinders precise gripper control, whereas Mobile ALOHA's physical base constrains operator movement. On *Push Chair*, this constraint leads the operator to pause before grasping, increasing demonstration time despite a high success rate. HALO addresses these difficulties through direct manipulation control and a low-resistance platform that allows the operator to reposition naturally.

4.2. Dexterity

We evaluate whether HALO can capture tasks requiring precise manipulator and base control, and whether the learned policy can reproduce high-precision behaviors.

Task description. As shown in Fig. 9, *Zip-Tie Fastening* requires a sequence of precise bimanual operations, including picking, handover, folding, insertion, fastening, and placement. The policy must manipulate a short, small zip tie throughout this process, requiring manipulation accuracy on the order of sub-millimeter.

Results. As shown in Fig. 9 and Tab. 2, the policy successfully reproduces the precise manipulation required for Zip-Tie Fastening. In contrast, Mobile ALOHA suffers from the perception gap, as the operator remains too far from the zip tie to clearly observe the fine manipulation details. VR teleoperation cannot reliably achieve the control precision and dexterity required for folding and fastening the zip tie. As a result, both interfaces yield low demonstration success rates and poor-quality data, preventing the learned policies from successfully completing the task.

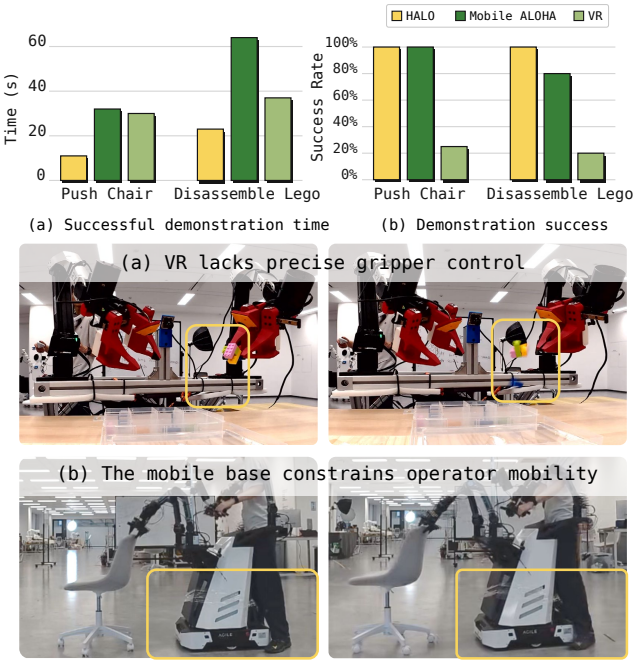


Figure 8: **Comparison and limitations of demonstration data collection. Top:** Comparison of human demonstration collection methods in terms of efficiency and reliability. Succ. Rate and Time denote the demonstration success rate and the average duration of successful demonstrations, respectively. **Bottom:** Representative limitations of existing teleoperation interfaces: VR lacks precise gripper control and force feedback, while the mobile base constrains operator mobility during whole-body demonstration collection.

Table 2: **Dexterity evaluation.** Zip-Tie Fastening is evaluated over 10 trials. Succ. Rate denotes the task success rate, and Time denotes the average completion time.

Zip-Tie Fastening					
HALO		ALOHA		VR	
Succ. Rate	Time (s)	Succ. Rate	Time (s)	Succ. Rate	Time (s)
30.0%	41	0.0%	-	0.0%	-

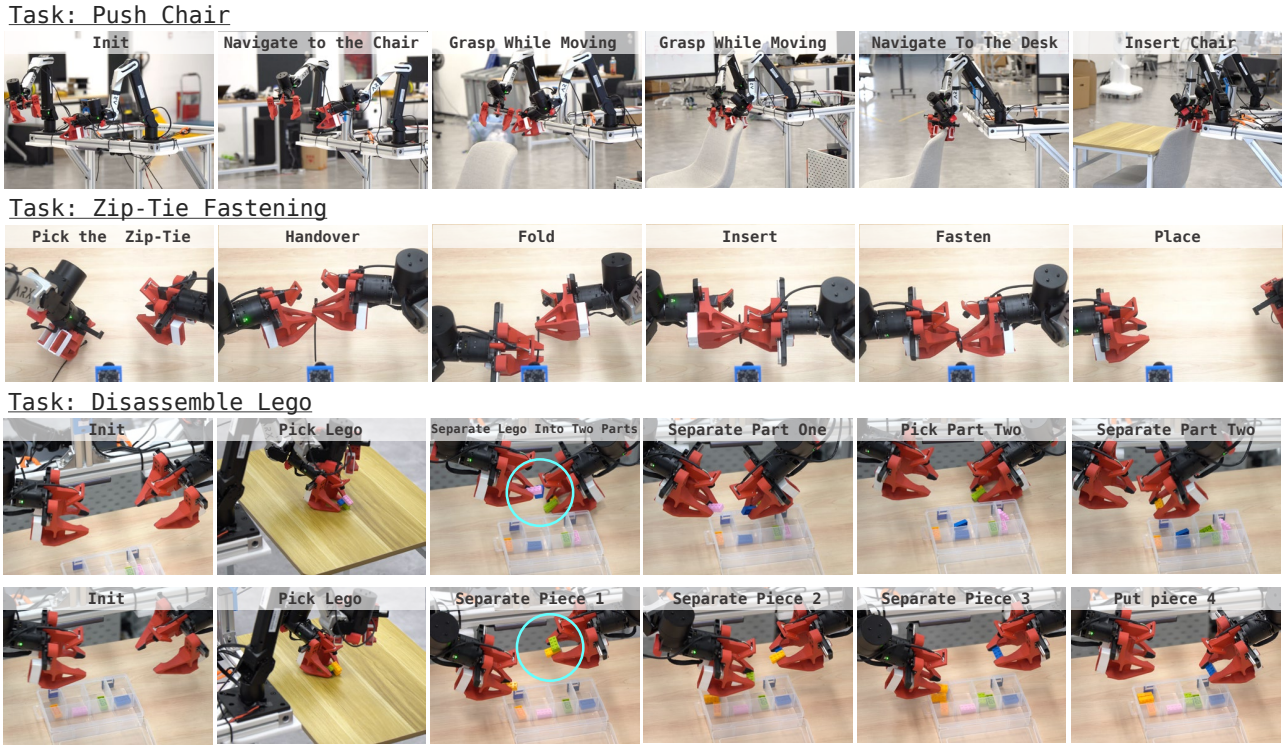


Figure 9: **Push Chair, Zip-Tie Fastening, and Disassemble Lego.** In Push Chair, the policy navigates to the chair, grasps it while moving, transports it to the target desk, and inserts it into place. Both Zip-Tie Fastening and Disassemble Lego require precise action prediction. In Disassemble Lego, the randomized separation pattern changes the required disassembly strategy. The policy learns to handle these variations from our demonstrations and generalizes to unseen Lego configurations. In the third row, two Lego pieces of the same color are included, and the policy correctly places both resulting pieces into the correct bin.

4.3. Coordinated whole-body control

We test HALO on tasks that require both precise whole-body coordination and long-horizon mobile manipulation.

Task description. We evaluate two tasks: In *Disassemble Lego*, the policy must precisely position the base relative to the workspace and coordinate both end-effectors to grasp, separate, and place small interlocked Lego pieces. Varying connection strengths require coordinated bimanual force and motion control together with accurate base positioning, as shown in Fig. 9.

For the long-horizon task, the procedure consists of six stages as shown in Fig. 10. The policy first retrieves a box from the shelf, then places target objects into the box. It next lifts the loaded box with both arms, navigates to the conveyor, and places the box at the designated location. Finally, the robot returns to the shelf and repeats the procedure. The task requires coordinated bimanual manipulation, accurate base navigation, and reliable long-horizon execution across an extended workspace.

Results. In *Disassemble Lego*, the policy succeeds in 30 out of 40 trials (75.0%). For the long-horizon box manipulation task, the policy succeeds in 11 out of 15 trials (73.3%). Despite multiple forward, backward, and rotational base motions throughout the task, the policy consistently returns the mobile base to its initial position, enabling the full procedure to be repeated reliably. For both VR and Mobile ALOHA, policies trained with the same data collection time budget fail on both tasks, due to slow demonstrations with frequent pauses and insufficient valid data resulting from high collection failure rates.

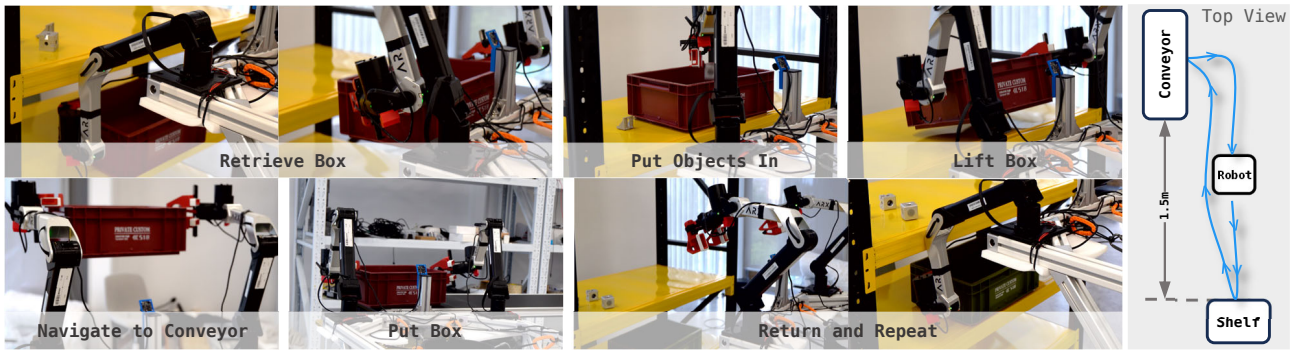


Figure 10: **Long-horizon mobile manipulation with coordinated whole-body control.** The policy retrieves a box from the shelf, loads target objects, transports the box to the conveyor, places it at the designated location, and returns to repeat the procedure.

Table 3: **Inference Strategy.** Success rates on the *Push Chair* task under synchronous and asynchronous inference with different execution horizons.

Push Chair	Synchronous		Asynchronous			
Execution Horizon	8	25	4	6	8	10
Success Rate	0.0%	35.0%	75.0%	75.0%	85.0%	70.0%

Table 4: **Action representation ablation.** Success rates of *Disassemble Lego* using different actions space and base model.

	From VLM	Pretrained VLA
EEF	12.5%	47.5%
Joint	7.5%	75.0%

4.4. Ablation Studies

Inference strategies. Since tight whole-body coordination is sensitive to inference pauses, we further study the effect of RTC-specific execution design on policy performance, as shown in Tab. 3. Under *synchronous execution*, inference pauses cannot be fully eliminated, and longer execution horizons yield higher success rates because frequent replanning introduces more pauses and disrupts coordination between the arms and mobile base. *Asynchronous execution* achieves higher overall success by mitigating the effect of inference latency. However, more frequent replanning does not always improve performance, indicating a trade-off between reactivity and temporal consistency. Frequent replanning can reduce temporal consistency across action chunks, while longer horizons reduce responsiveness to new observations. Our results show that an intermediate replanning horizon provides the best balance.

Deployment-aligned visual post-processing. We evaluate whether removing demonstrator-specific visual cues improves alignment with autonomous deployment. On *Push Chair*, we compare policies trained on the same demonstrations with and without the visual processing in Section 3.3.1, using the best inference setting. Hand removal improves the success rate from 55% to 85%, a gain of 30 percentage points, demonstrating the effectiveness of deployment-aligned visual processing.

Action representation. We compare end-effector (EEF) and joint-space actions. As shown in Tab. 4, EEF performs better from scratch, while joint-space actions achieve higher success after pretraining. This suggests that pretraining enables the policy to better exploit direct joint-space control.

4.5. Generalization of mobile manipulation.

Although the target table remains fixed during data collection, the learned VLA policy generalizes *Push Chair* to unseen backgrounds and substantially different chair–table configurations, as shown in Fig. 11. We attribute this generalization partly to velocity-based base control, which encourages the policy to learn a reactive motion pattern that continuously drives the chair toward the table rather than memorizing a fixed global pose.

5. Conclusion

In this work, we presented HALO, a human-aligned data collection system for whole-body mobile manipulation. Our key insight is that the robot interface should adapt to human capabilities by aligning perception and manipulation while preserving natural locomotion. This design enables efficient collection of coordinated demonstrations, while demonstrator decoupling mitigates the visual and control shifts introduced by close human–robot coupling. Building on this alignment, future work could incorporate active perception (Xiong et al., 2025; Xu et al., 2026) to adapt the robot viewpoint during manipulation and more closely reflect human visual behavior. Across rapid mobile manipulation, dexterous tasks, and long-horizon behaviors, HALO demonstrates that human-aligned interfaces provide an effective foundation for efficient data collection and robust whole-body policy learning.

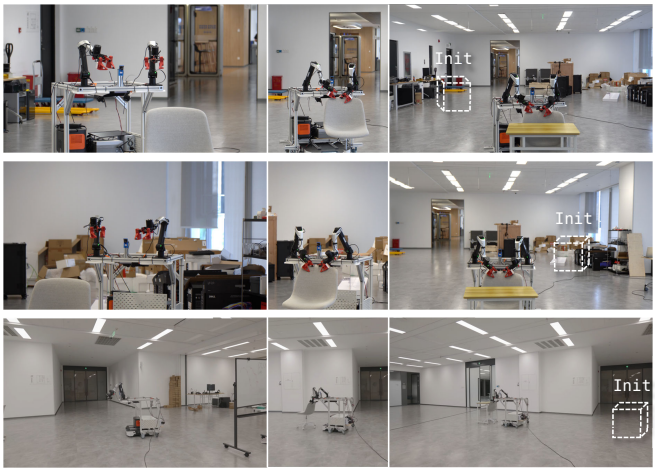


Figure 11: **Generalization of mobile manipulation.** We evaluate *Push Chair* under large initial-pose variations and in unseen environments. Compared with the 5–10 cm initial-position variation in Mobile ALOHA, we increase the randomization to 2–3 m.

References

Akgun, B., Cakmak, M., Yoo, J. W., and Thomaz, A. L. Trajectories and keyframes for kinesthetic teaching: A human-robot interaction perspective. In *Proceedings of the seventh annual ACM/IEEE international conference on Human-Robot Interaction*, pp. 391–398, 2012. 2

Black, K., Brown, N., Driess, D., Esmail, A., Equi, M., Finn, C., Fusai, N., Groom, L., Hausman, K., Ichter, B., et al. π_0: A vision-language-action flow model for general robot control. *arXiv preprint arXiv:2410.24164*, 2024. 1

Black, K., Ren, A. Z., Equi, M., and Levine, S. Training-time action conditioning for efficient real-time chunking. *arXiv preprint arXiv:2512.05964*, 2025. 7

Chen, T., Wu, H., Wang, J., Li, X., Jin, Z., and Fang, L. Supersuit: An isomorphic bimodal interface for scalable mobile manipulation. *arXiv preprint arXiv:2603.06280*, 2026. 3

Chi, C., Xu, Z., Pan, C., Cousineau, E., Burchfiel, B., Feng, S., Tedrake, R., and Song, S. Universal manipulation interface: In-the-wild robot teaching without in-the-wild robots. *arXiv preprint arXiv:2402.10329*, 2024. 2, 3

Chi, C., Xu, Z., Feng, S., Cousineau, E., Du, Y., Burchfiel, B., Tedrake, R., and Song, S. Diffusion policy: Visuomotor policy learning via action diffusion. *The International Journal of Robotics Research*, 44 (10-11):1684–1704, 2025. 1

Citron, J., Zbizika, R., Liu, Z., and Song, S. Modpack: An extensible teleoperation interface for bimanual mobile manipulation. *arXiv preprint arXiv:2607.19479*, 2026. 3

Fu, Z., Zhao, T. Z., and Finn, C. Mobile aloha: Learning bimanual mobile manipulation with low-cost whole-body teleoperation. *arXiv preprint arXiv:2401.02117*, 2024. 1, 3, 4, 5, 6

Ha, H., Gao, Y., Fu, Z., Tan, J., and Song, S. Umi on legs: Making manipulation policies mobile with manipulation-centric whole-body controllers. *arXiv preprint arXiv:2407.10353*, 2024. doi: 10.48550/arXiv.2407.10353. 4

Honerkamp, D., Mahesheka, H., von Hartz, J. O., Welschehold, T., and Valada, A. Whole-body teleoperation for mobile manipulation at zero added cost. *IEEE Robotics and Automation Letters*, 10 (4):3198–3205, 2025. 1, 3

Intelligence, P., Black, K., Brown, N., Darpinian, J., Dhabalia, K., Driess, D., Esmail, A., Equi, M., Finn, C., Fusai, N., et al. $\pi_{0.5}$: a vision-language-action model with open-world generalization. *arXiv preprint arXiv:2504.16054*, 2025. 4, 7

Jiang, Y., Zhang, R., Wong, J., Wang, C., Ze, Y., Yin, H., Gokmen, C., Song, S., Wu, J., and Fei-Fei, L. Behavior robot suite: Streamlining real-world whole-body manipulation for everyday household activities. *arXiv preprint arXiv:2503.05652*, 2025. 3

Kim, M. J., Pertsch, K., Karamcheti, S., Xiao, T., Balakrishna, A., Nair, S., Rafailov, R., Foster, E., Lam, G., Sanketi, P., et al. Openvla: An open-source vision-language-action model. *arXiv preprint arXiv:2406.09246*, 2024. 1

Li, L., Zhang, Q., Luo, Y., Yang, S., Wang, R., Han, F., Yu, M., Gao, Z., Xue, N., Zhu, X., et al. Causal world modeling for robot control. *arXiv preprint arXiv:2601.21998*, 2026. 1

Li, X., Xue, H., Ren, P., and Bo, L. Diffueraser: A diffusion model for video inpainting. *arXiv preprint arXiv:2501.10018*, 2025. 6

Liang, L., Xu, J., Qi, X., Cai, Y., Ding, H., Wang, L., Sun, Z., Chow, J.-H., Yang, M., and Cutkosky, M. Universal manipulation exoskeleton: Learning compliant whole-body policies with real-time torque feedback. *arXiv preprint arXiv:2606.14218*, 2026. 3

Lum, T. G. W., Lee, O. Y., Liu, C. K., and Bohg, J. Crossing the human-robot embodiment gap with sim-to-real rl using one human demonstration. *arXiv preprint arXiv:2504.12609*, 2025. 4

Punamiya, R., Patel, D., Aphiwetsa, P., Kuppili, P., Zhu, L. Y., Kareer, S., Hoffman, J., and Xu, D. Egobridge: Domain adaptation for generalizable imitation from egocentric human data. In *Human to Robot: Workshop on Sensorizing, Modeling, and Learning from Humans*, 2025. 4

Rakita, D., Mutlu, B., and Gleicher, M. An autonomous dynamic camera method for effective remote teleoperation. In *Proceedings of the 2018 ACM/IEEE International Conference on Human-Robot Interaction*, pp. 325–333, 2018. 1, 3

Ravi, N., Gabeur, V., Hu, Y.-T., Hu, R., Ryali, C., Ma, T., Khedr, H., Rädle, R., Rolland, C., Gustafson, L., et al. Sam 2: Segment anything in images and videos. In *International Conference on Learning Representations*, volume 2025, pp. 28085–28128, 2025. 6

Su, Y., Wang, Y., and Chau, L.-P. Care-ego: Contact-aware relationship modeling for egocentric interactive hand-object segmentation. *Expert Systems with Applications*, pp. 129148, 2025. 6

Wang, Y., Lin, P., Chen, X.-H., Yuan, H., Liang, Z., Huang, Y., Chen, A., Lei, Z., Zhang, J., Zhang, T., et al. Ego2robot: Scalable robot data synthesis from egocentric human data. *arXiv preprint arXiv:2608.02580*, 2026. 4

Wong, J., Tung, A., Kurenkov, A., Mandlekar, A., Fei-Fei, L., Savarese, S., and Martín-Martín, R. Error-aware imitation learning from teleoperation data for mobile manipulation. In *Conference on Robot Learning*, pp. 1367–1378. PMLR, 2022. 3

Wu, J., Chong, W., Holmberg, R., Prasad, A., Gao, Y., Khatib, O., Song, S., Rusinkiewicz, S., and Bohg, J. Tidybot++: An open-source holonomic mobile manipulator for robot learning. *arXiv preprint arXiv:2412.10447*, 2024a. 4

Wu, P., Shentu, Y., Yi, Z., Lin, X., and Abbeel, P. Gello: A general, low-cost, and intuitive teleoperation framework for robot manipulators. In *2024 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pp. 12156–12163. IEEE, 2024b. 3

Wu, W., Lu, F., Wang, Y., Yang, S., Liu, S., Wang, F., Zhu, Q., Sun, H., Wang, Y., Ma, S., et al. A pragmatic vla foundation model. *arXiv preprint arXiv:2601.18692*, 2026. 1

Xiong, H., Xu, X., Wu, J., Hou, Y., Bohg, J., and Song, S. Vision in action: Learning active perception from human demonstrations. *arXiv preprint arXiv:2506.15666*, 2025. 11

Xu, M., Zhang, H., Hou, Y., Xu, Z., Fan, L., Veloso, M., and Song, S. Dexumi: Using human hand as the universal manipulation interface for dexterous manipulation. *arXiv preprint arXiv:2505.21864*, 2025. 3

Xu, X., Park, J., Zhang, H., Cousineau, E., Bhat, A., Barreiros, J., Wang, D., Bohg, J., and Song, S. Hommi: Learning whole-body mobile manipulation from human demonstrations. *arXiv preprint arXiv:2603.03243*, 2026. 2, 3, 4, 11

Yu, J., Shentu, Y., Wu, D., Abbeel, P., Goldberg, K., and Wu, P. Egomi: Learning active vision and whole-body manipulation from egocentric human demonstrations. *arXiv preprint arXiv:2511.00153*, 2025. 3

Yuan, Z., Wei, T., Gu, L., Hua, P., Liang, T., Chen, Y., and Xu, H. Hermes: Human-to-robot embodied learning from multi-source motion data for mobile dexterous manipulation. *IEEE Transactions on Robotics*, 2026. 4

Zhao, J., Tassi, F., Huang, Y., De Momi, E., and Ajoudani, A. A combined learning and optimization framework to transfer human whole-body loco-manipulation skills to mobile manipulators. *Robotics and Autonomous Systems*, 189:104958, 2025. 4

Zhao, T. Z., Kumar, V., Levine, S., and Finn, C. Learning fine-grained bimanual manipulation with low-cost hardware. *arXiv preprint arXiv:2304.13705*, 2023. 1, 3

Zhu, L. Y., Kuppili, P., Punamiya, R., Aphiwetsa, P., Patel, D., Kareer, S., Ha, S., and Xu, D. Emma: Scaling mobile manipulation via egocentric human data. *IEEE Robotics and Automation Letters*, 2026. 4

Zhu, Y., Stone, P., and Zhu, Y. Bottom-up skill discovery from unsegmented demonstrations for long-horizon robot manipulation. *IEEE Robotics and Automation Letters*, 7(2):4126–4133, 2022. 1, 2, 3